

Windows and WSL2

Using WSL2 on Windows

- This page explains how to use WSL2 (Windows Subsystem for Linux 2).
- All ASE 230 shell scripts and Linux command-line tools are written for Linux. **Windows users must run everything inside WSL2.**
- If you are using Linux or macOS, you can skip these slides.

1. Install WSL2 and Ubuntu

The easiest method works on Windows 11 and Windows 10 version 2004 or later.

1. Search for **PowerShell**, right-click it, and select **Run as administrator**.

2. Run this one command:

```
wsl --install -d Ubuntu-24.04
```

This enables WSL, installs WSL2, and installs Ubuntu 24.04.

- The `-d Ubuntu-24.04` option gives every ASE 230 student the same predictable environment.
- Plain `wsl --install` installs the current default distribution (normally Ubuntu), but the exact Ubuntu release selected by that default can change over time.

Finish the Ubuntu Setup

Restart Windows if prompted.

1. After restarting, open **Ubuntu** from the Start menu or Windows Terminal.
2. Wait while Ubuntu finishes installing.
3. Create a Linux username and password when prompted.

Nothing appears while you type a Linux password. This is normal - type it and press Enter.

To see the distribution names currently available, run:

```
wsl --list --online
```

- Official guide:
<https://learn.microsoft.com/windows/wsl/install>
- If necessary, use the last section of this document to uninstall/reinstall WSL2.

2. Open the WSL2 (Ubuntu) Terminal

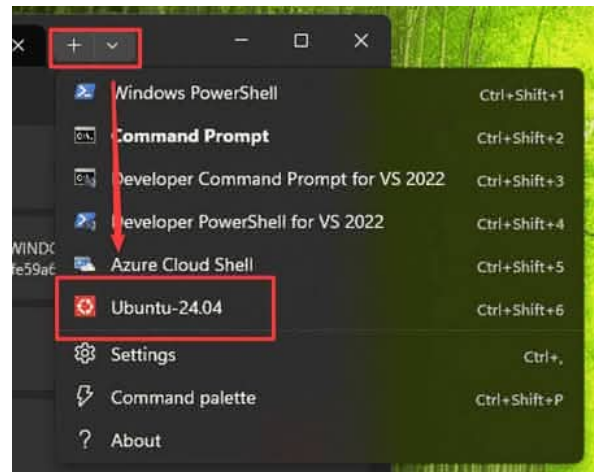
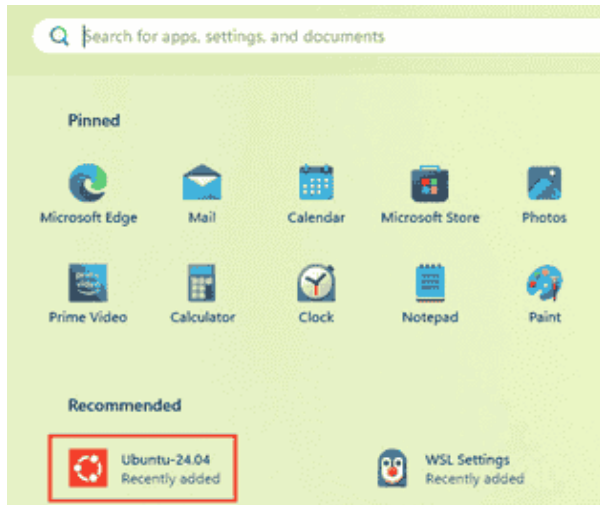
Windows Terminal (Recommended)

Windows Terminal is recommended because it provides convenient tabs for PowerShell, Command Prompt, and installed WSL distributions.

- It is normally pre-installed on Windows 11.
- On Windows 10, if it is not installed, open the **Microsoft Store**, search for **Windows Terminal**, and select **Install**.

Open Ubuntu-24.04

Open Ubuntu-24.04 from the Start menu (left) or open an Ubuntu-24.04 tab in Windows Terminal (right).



3. Update Ubuntu

Then update Ubuntu:

```
sudo apt update  
sudo apt upgrade -y
```

4. Verify WSL2 Is Set Up Correctly

Open **PowerShell on Windows** - not the Ubuntu terminal - and run:

```
wsl --list --verbose
```

Confirm that Ubuntu's **VERSION** column shows **2**.

```
PS C:\WINDOWS\system32> wsl --list --verbose
  NAME                STATE      VERSION
* Ubuntu-24.04        Running    2
```

5. Accessing Files

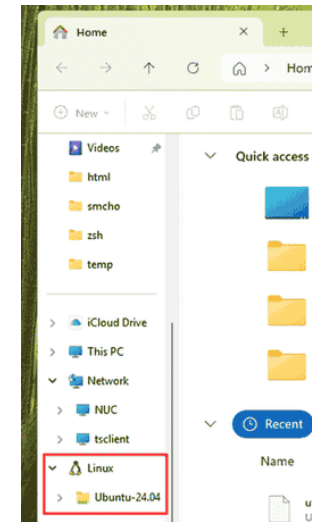
Accessing Windows Files from WSL2

- The `C:` drive is mounted as `/mnt/c` inside WSL2.

```
# C:\temp\ex on Windows is the same as:  
cd /mnt/c/temp/ex
```

Accessing WSL2 from Windows

- The WSL2/Ubuntu directories can be accessed from Explorer.
- From Windows, WSL2 behaves like a separate machine on the network, which is why Explorer treats it as a network location.



6. Line-Ending (EOL) Errors

Windows and Linux use different line endings. A script edited on Windows may fail like this:

```
./setup.sh: line 5: $'\r': command not found
```

Fix it with `dos2unix`:

```
sudo apt install dos2unix  
dos2unix ./setup.sh
```

Next: Install the Course Tools

After `wsl --list --verbose` confirms that Ubuntu is using WSL2, continue with:

1. `03. php installation.pdf`
2. `04. mysql installation.pdf`

Windows users should run the **Linux / WSL2 (Ubuntu)** commands in those guides inside the Ubuntu terminal.

Getting Unstuck

Configurations and repositories are updated frequently, so installation steps may occasionally cause errors. If that happens, you can:

- Copy and paste the error message into an LLM (Claude, ChatGPT, Perplexity, etc.) to get a quick fix.
- Email me the error details.
- Visit me in person so we can troubleshoot together.

(Optional) Uninstall/Reinstall WSL2

There are two different cleanup options. Open **PowerShell as administrator** before using either one.

Warning: `wsl --unregister <DistroName>` permanently deletes that Linux distribution, including its files, installed programs, and settings. Copy any work you need to keep to Windows or another backup first.

Option A: Reset Ubuntu Only

Use this when Ubuntu is broken and you want a fresh classroom setup, but you want to keep WSL2 installed.

First, see the exact distribution names installed on the computer:

```
wsl --list --verbose
```

Stop WSL, then unregister only the distribution you want to reset. Replace `Ubuntu` if the list shows a different name, such as `Ubuntu-24.04`.

```
wsl --shutdown  
wsl --unregister Ubuntu
```

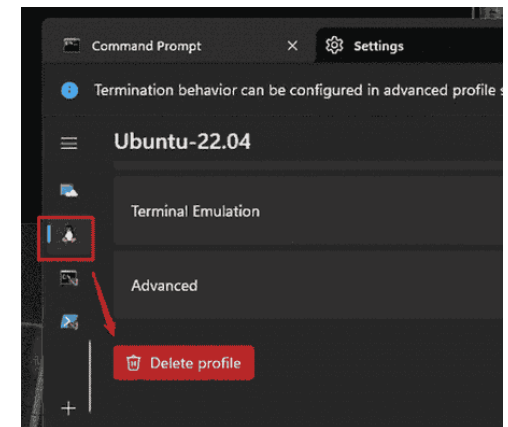
Windows Terminal

After unregistering, run `wsl --list --verbose` again.

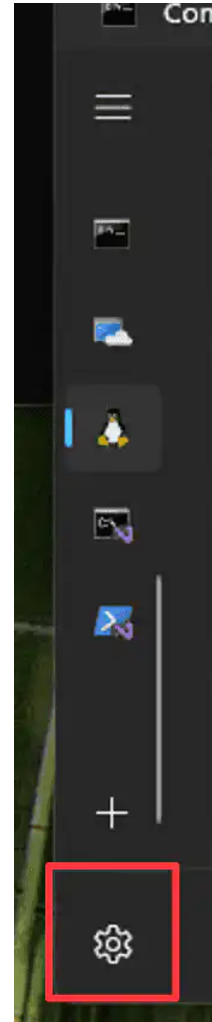
- If the distribution is no longer listed, it has been removed from WSL even if Windows Terminal still shows its old profile. Clicking that stale profile may display `WSL_E_DISTRO_NOT_FOUND`.

Windows Terminal normally creates WSL profiles dynamically and refreshes them when it restarts.

- Completely close all Windows Terminal windows and processes, then reopen it.
- If the stale entry remains, open **Settings** and remove any manually created profile for the deleted distribution.



- You can also use **Open JSON file** to inspect `settings.json` for a manually persisted profile.
- A stale Terminal profile does not mean WSL or Ubuntu must be reinstalled.



Reinstall a fresh copy when you are ready:

```
wsl --install -d Ubuntu-24.04
```

Option B: Remove WSL2 Completely

Use this only when you no longer want WSL or any Linux distributions on the computer.

1. Back up any Linux files you need.
2. Run `wsl --list --verbose` and unregister every listed distribution you want to delete.
3. Try the modern uninstall command:

```
wsl --uninstall
```

Automated WSL2 Cleanup

If `wsl --uninstall` is not supported on the computer, use the script below. It can unregister the selected distributions and disable the Windows features used by WSL2.

```
Set-ExecutionPolicy -Scope Process Bypass  
& ".\uninstall_wsl2.ps1"
```


The script is located in:

```
course_tools/code/02. install_ase230_tools_with_wsl2/uninstall_wsl2.ps1
```

Choose **Reset one distro** to keep WSL2 and delete only one Linux installation.

- Choose **Remove WSL2 completely** to unregister every installed distribution and disable the WSL-related Windows optional features.
- Disabling **Virtual Machine Platform** may also affect software such as Docker Desktop, so use the complete-removal option only on a computer that no longer needs it. Restart Windows when prompted.

Verify Complete Removal

After a complete removal and restart, these commands should show no installed distributions or report that WSL is not installed:

```
wsl --list --verbose  
wsl --status
```